

Flutter

Cross platform for building Mobile Apps

Summer Training • Academic Year 2025 – 2026

Prepared with : Eng.Hadi Heikal



Course Agenda



01

Session 1: Setup, Variables & Control Flow

- Introduction to Flutter & Dart
- Tools : VS_Code, Git, GitHub, Figma
- Set up Dart & VS Code
- Data Type & Variable, Naming conventions
- var v.s dynamic
- const v.s final
- User Input
- Conditional Statements
 - (if-else), switch cases, Ternary Operator
- Loops (While Loop, Do While Loop, For Loop)



Course Agenda



02

Session 2: Collections, Functions & Error Handling

- Parse and cast
- List, Set, Map
- Dart Functions
 - Types Of Function
 - Function Parameter In Dart
 - Anonymous Function In Dart
 - arrow function
- Exception Handling in Dart (Try / Catch)
- Late Keyword In Dart
- Null Safety
- Types of errors



Course Agenda



03

Session 3: Object-Oriented Programming (OOP)

- Intro to OOP, Classes & Objects
- Constructors
- Default, Parameterized, Named, Constant Constructor
- Encapsulation (Getters, Setters)
- Inheritance, Inheritance Of Constructor, Super
- Polymorphism
- Static, Enum
- Abstraction
- Interface, Mixin in Dart
- Factory Constructor
- Generic



Course Agenda



04

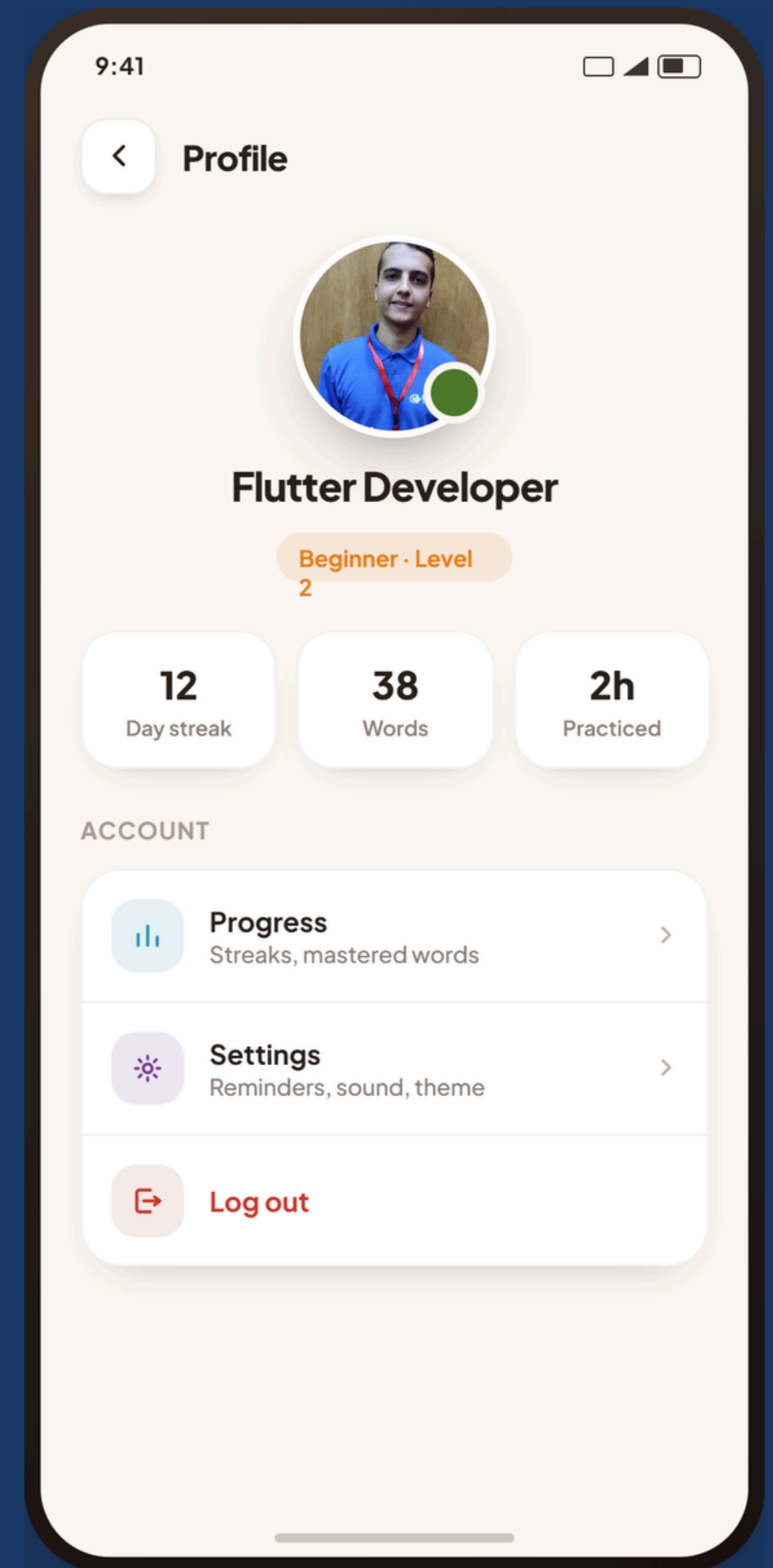
Session 4: Flutter Intro, Widget Tree & Basic UI

- install Flutter SDKs & Plugins, Emulator
- Intro to Flutter & Flutter Developer Role
- Architecture of Flutter & Widget Tree
- Stateless vs Stateful Widgets (lifecycle)
- Run the first application in emulator
- Material App
- Basic UI Widgets (Scaffold, AppBar, Text, Icon, Container, Padding, Margin, Center, Image, Fonts)
- Layout Widgets (Row, Column, Stack, Positioned)
- Space Management (Expanded, Flexible, SizedBox)
- **Hands-on: Building a modern Language Learning App**
 - **Profile / Splash Screen**



Splash Screen

Profile Screen



Course Agenda



05

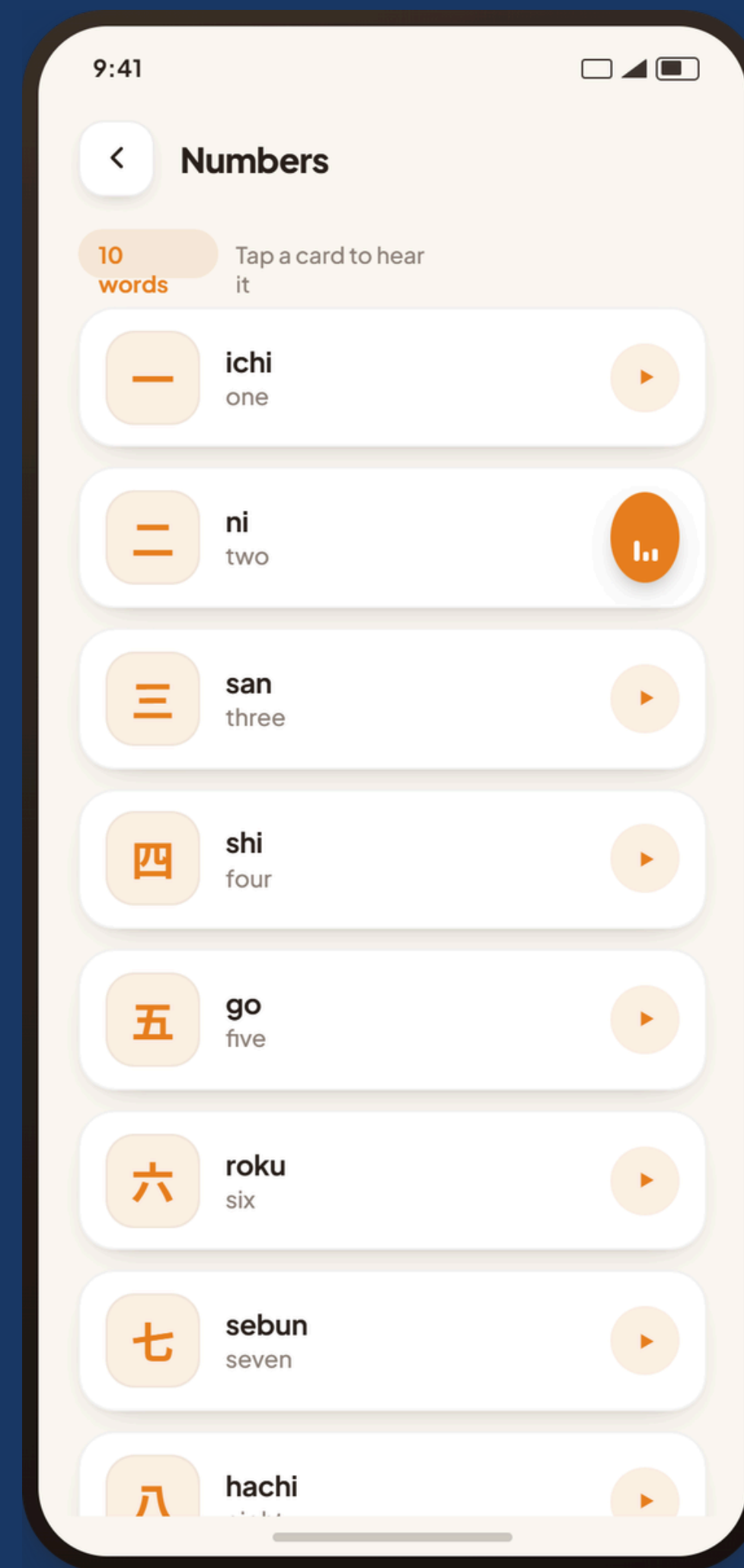
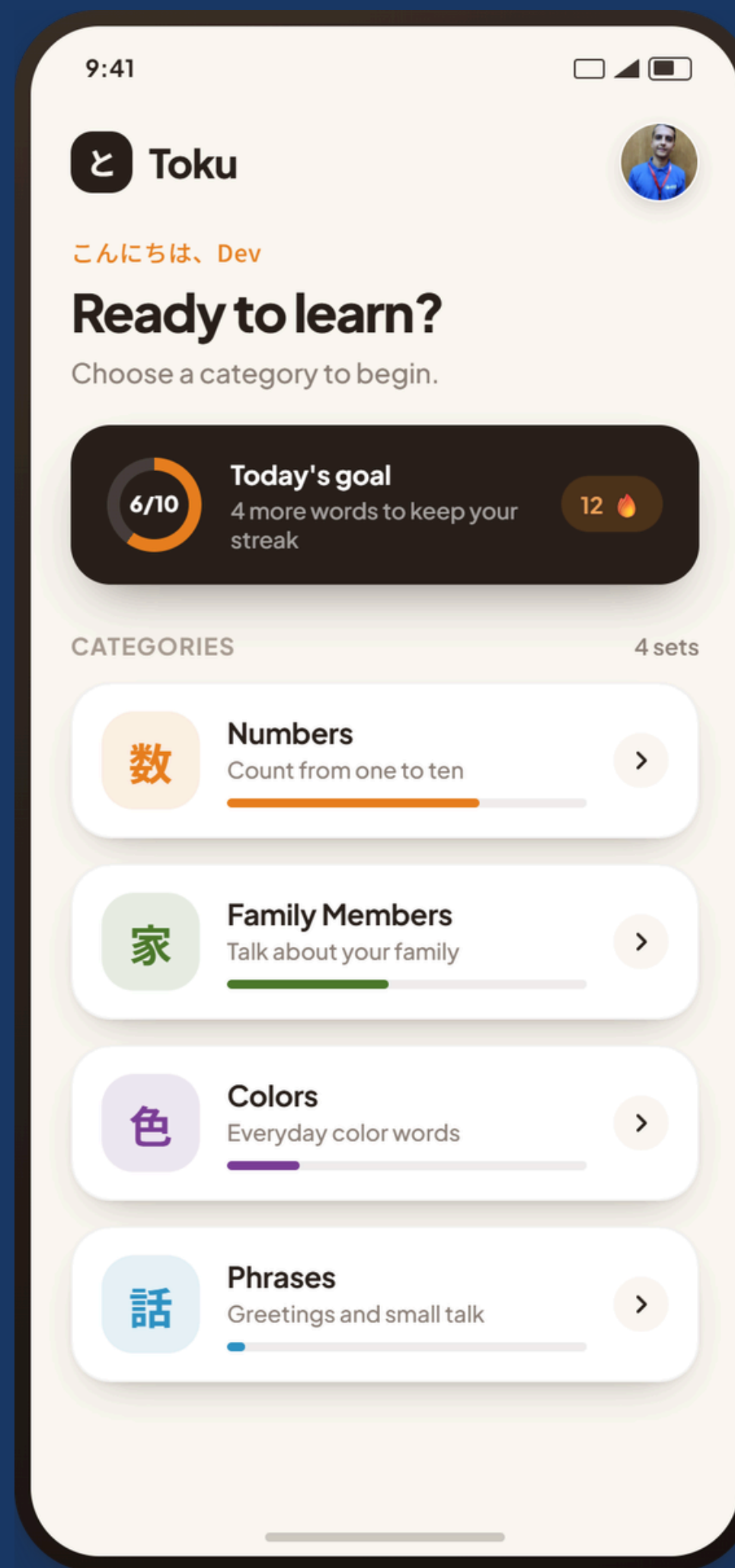
Session 5: Building Smart UIs (Lists & Navigation)

- SafeArea & SingleChildScrollView
- Modern UI Widgets: Card, CircleAvatar, and IconButton
- Interactivity: GestureDetector & InkWell
- Custom Widgets: Code extraction & reusability
- Data Models: Structuring dynamic data using basic OOP classes
- Lists: ListView & ListView.builder for efficient dynamic data rendering
- Navigation Basics: Navigator.push (Connecting screens)
- External Packages: Installing libraries & playing audio (e.g., audioplayers)
- **Hands-on: Building a complete Language Learning App**
 - (Home, Category details , and playing sounds)



Home Screen

category details Screen



Course Agenda



06

Session 6: Local Storage & Firebase Auth

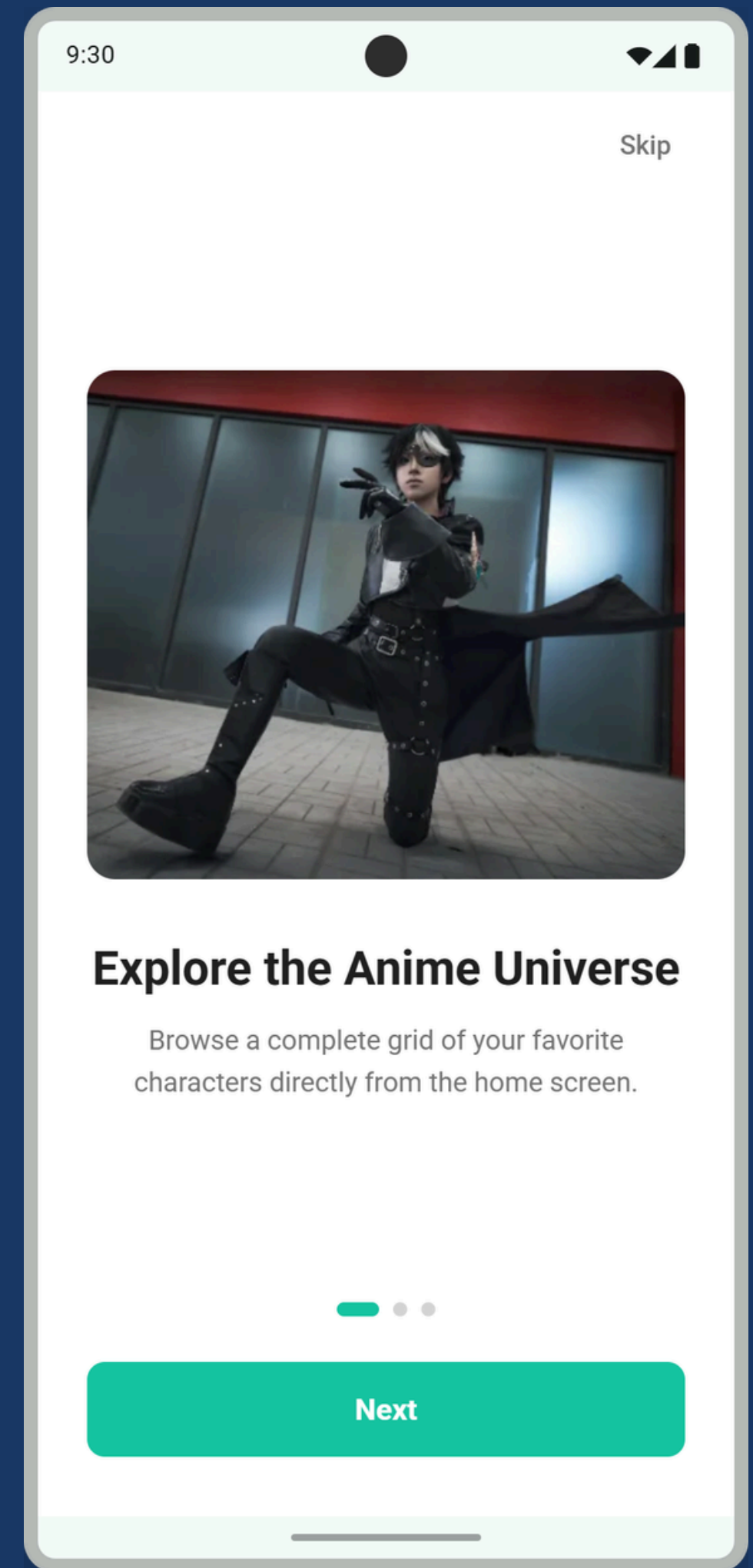
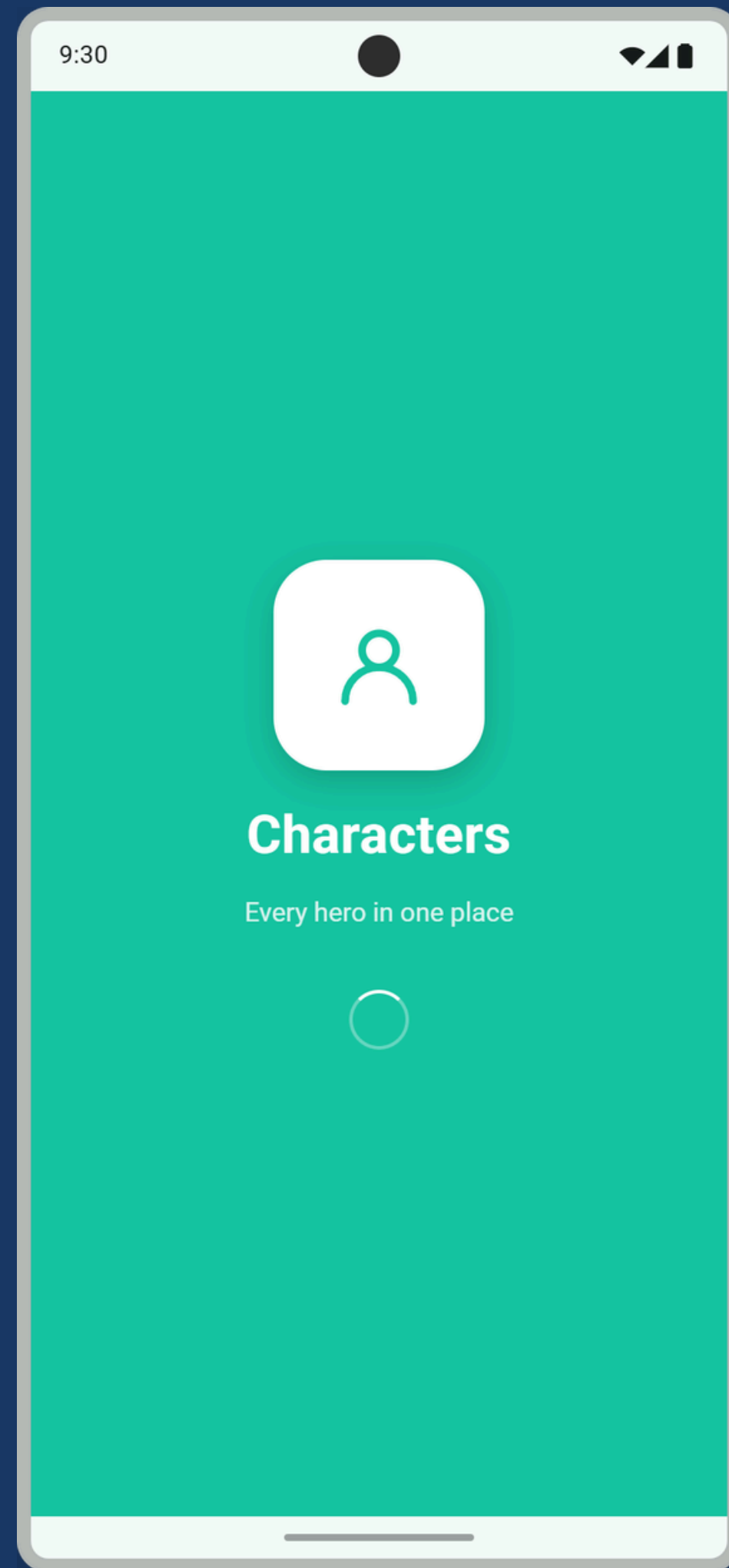
- Project Setup (System Design/Intro to Architecture Patterns)
- Navigation
- Splash screen (using flutter native splash package)
- Onboarding Screens
- SharedPreferences: Managing the "First-Time User" experience (showing onboarding only once).
- Authentication UI: Building responsive Login and Registration forms with validation.
- Firebase Integration: Setting up the project and implementing Email/Password Authentication.
- **Hands-on: Building a complete Anime Characters App**
 - **splash / onboarding /Auth(register-login) Screens**



Splash Screen

Onboarding Screens

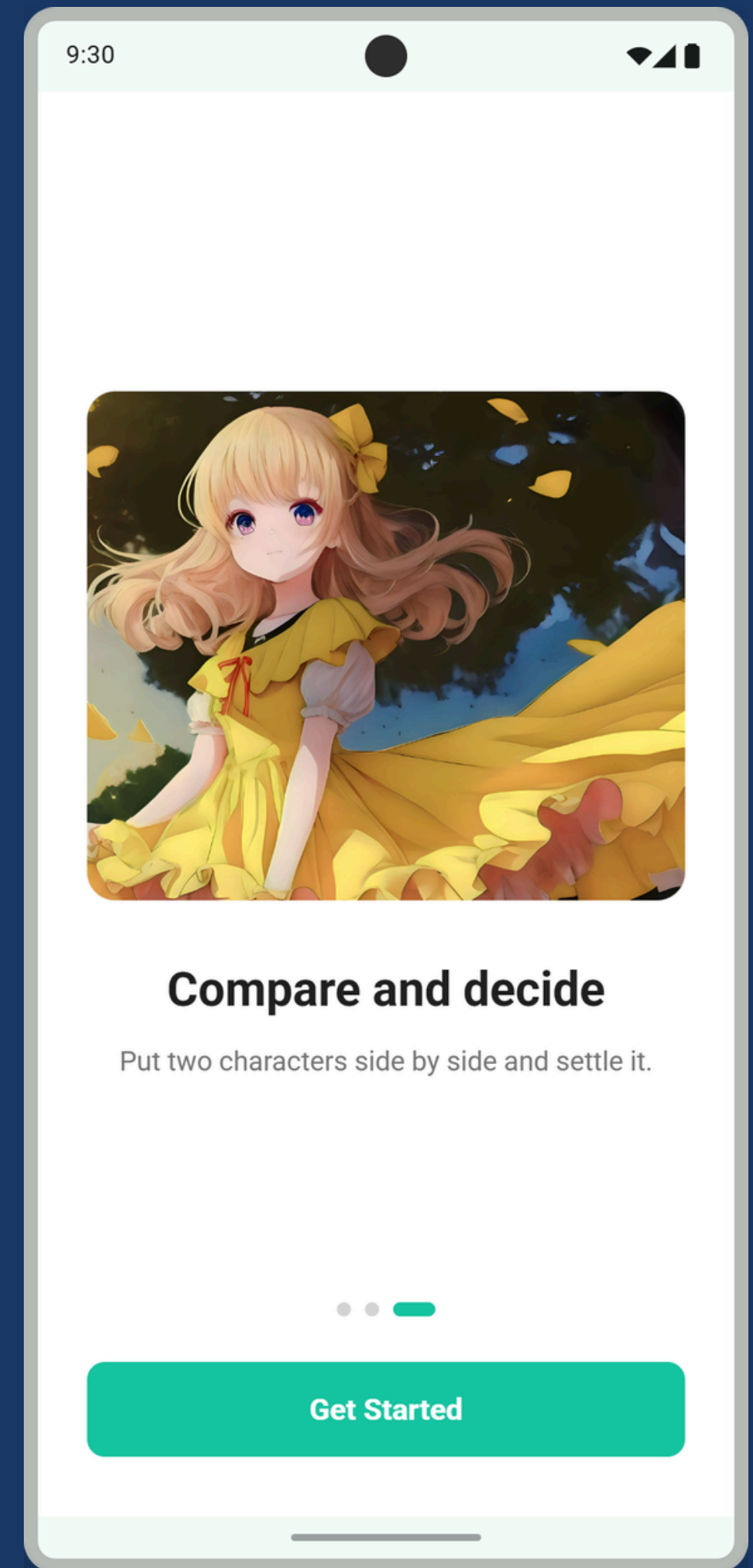
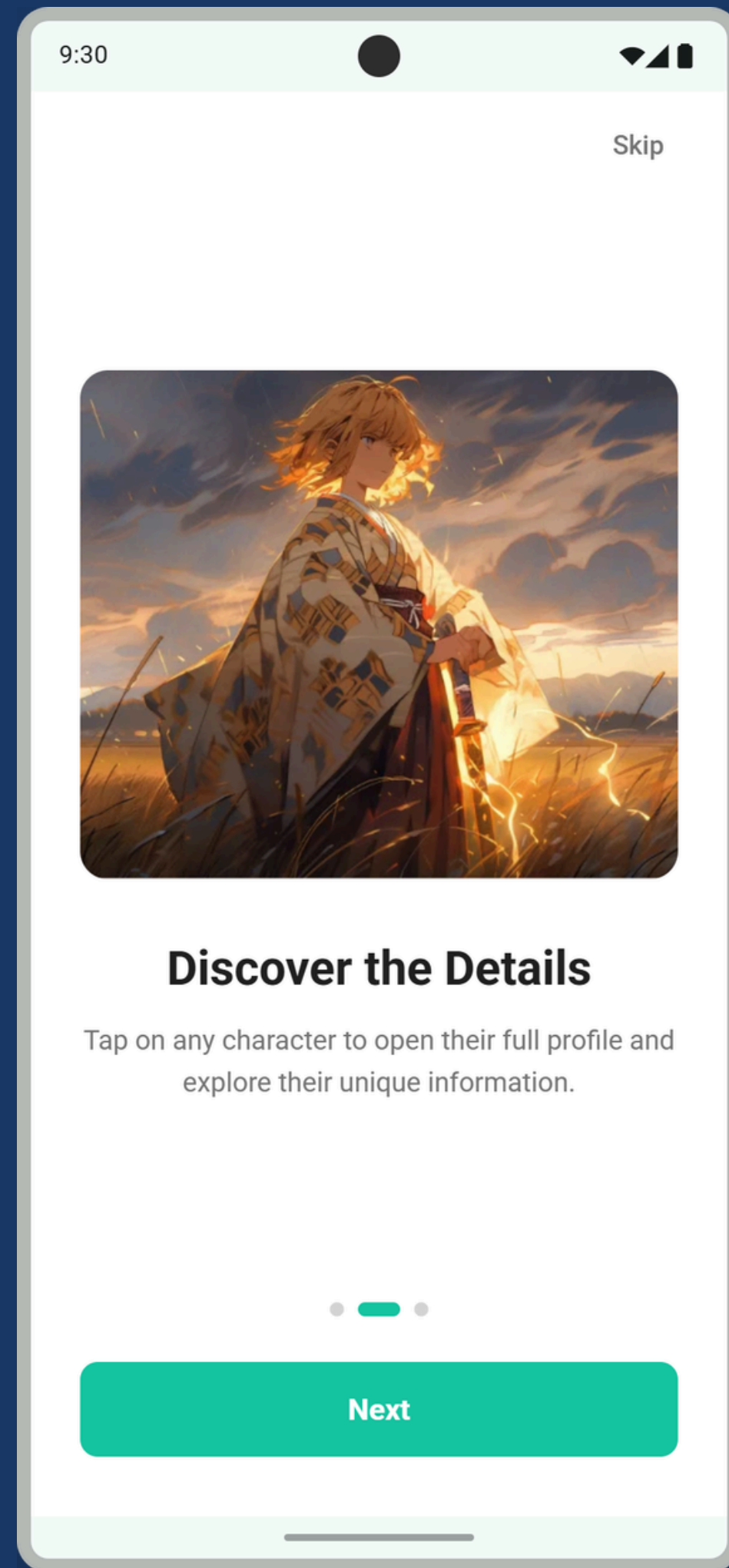
Auth Screens



Splash Screen

Onboarding Screens

Auth Screens



Splash Screen

Onboarding Screens

Auth Screens



9:30

Sign Up

Create an account

It takes about thirty seconds.

[Sign Up](#)

Already have an account? [Login](#)

9:30

Login

Welcome back

Log in to see all your characters.

[Login](#)

Don't have an account? [Sign up](#)

Course Agenda



07

Session 7: The Internet & APIs

- Home Screen UI: Building dynamic GridViews and ListViews (starting with static UI).
- API & JSON Fundamentals: Understanding RESTful APIs and how mobile apps communicate with servers.
- Fetching Data: Using the Dio package to retrieve character data from the Rick & Morty API.
- Data Models: Parsing JSON responses into structured Dart OOP classes.
- FutureBuilder: Handling network states seamlessly (Loading ↺, Success ✓, Error ✗).
- Hands-on: Building a complete Anime Characters App
 - Home Screen



Home Screens



Course Agenda



08

Session 08 : State Managemet & AI Power

- Character Details Screen: Passing object data between screens to show full character profiles.
- State Management (BLoC / Cubit): A foundational introduction to the Cubit/BLoC pattern, Refactoring our code
- The industry standard for separating business logic from the UI and managing state professionally.
- AI Tips & Tricks: Utilizing AI tools (like ChatGPT or Claude) to generate boilerplate code, debug errors, and enhance developer productivity.
- Launch: Building and exporting the final Android package (flutter build apk).
- Hands-on: Building a complete Anime Characters App
 - Characters Details Screen



Characters Details Screen

